

RAMANA KODURI

Full-Stack Software Developer | AI Application Engineering

Toronto, Canada • (514) 586-9903 • knvramana234@gmail.com • [LinkedIn](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 4+ years of experience building enterprise-grade applications and REST APIs across Python, Java, JavaScript/TypeScript, React, Django, and Spring Boot. Currently delivering production features for IBM Engineering Lifecycle Management (ELM) and building agentic AI prototypes that integrate Large Language Models via Model Context Protocol (MCP) servers using tool-based retrieval to ground natural-language responses in live enterprise data. Experienced in Python backend development (Django, Pandas, NumPy, Scikit-learn), Azure cloud services (Microsoft Certified: Azure Data Engineer Associate), and informal technical mentoring and code review. Seeking an Application Developer role building generative AI and retrieval-grounded solutions into enterprise products.

CORE COMPETENCIES

Full-Stack Development • Python / Django • Generative AI & Retrieval-Grounded Tool Calling • Agentic AI Prototyping • MCP Servers • Prompt Engineering • REST / OSLC API Design • React & TypeScript • Java / Spring Boot • Azure Cloud • Microservices • Code Reviews & Mentoring • System Design

PROFESSIONAL EXPERIENCE

IBM Canada — Software Developer

Jan 2025 – Present

Engineering Lifecycle Management (ELM)

- Built a full-stack agentic AI prototype integrating Large Language Models (IBM watsonx Granite) into the RMM web UI, using Model Context Protocol (MCP) tool calls to retrieve and ground responses in live RMM/ELM data spanning React/Dojo frontend, backend orchestration, prompt design, and end-to-end request/response handling; demoed to engineering leadership.
- Extended and modified Model Context Protocol (MCP) servers to expose RMM/ELM capabilities as agent-callable tools, enabling natural-language, retrieval-grounded interaction with structured enterprise data, services, and workflows.
- Designed and iterated on prompts, tool schemas, and orchestration logic for AI-assisted workflow generation evaluating model outputs, refining system prompts, sequencing tool calls, and validating responses against domain constraints.
- Provided informal code reviews and technical mentoring for teammates, including walkthroughs of the MCP/agentic AI prototype architecture to support broader team adoption of AI-assisted development patterns.
- Engineered full-stack enterprise features across RMM and IBM ELM, REST/OSLC service layers, Dojo-based web UI workflows, persistent data models, and cross-application integration flows used by Fortune 500 engineering teams.
- Coordinated iFix release readiness for IBM ELM interim-fix deliveries, tracking 15+ candidate defects across validation, backport decisions, and release scope to ensure on-time delivery to enterprise customers.
- Resolved 15+ customer-reported production defects spanning indexing, linking, permissions, configuration contexts, and cross-application integration, improving platform reliability and reducing recurring validation issues.
- Increased regression coverage by 15–20% for impacted backend-service and integration components using JUnit and Mockito, strengthening defect validation before iFix and final-build gates.
- Accelerated root-cause analysis for complex enterprise issues by correlating server logs, REST/OSLC traces, database records, validation reports, and HAR files across frontend, backend, persistence, and integration layers.

CognitiveBotics — Software Developer

Mar 2022 – Aug 2022

Ed-Tech Platform for Special-Needs Education • Hyderabad, India

- Delivered end-to-end full-stack features for an ed-tech platform supporting special-needs education across React, Python/Django REST APIs, and MongoDB including a real-time dashboard giving educators live visibility into student learning activity via WebSocket-driven updates.
- Cut dashboard response time by ~35% through Redis caching, query optimization, lazy loading, code splitting, and API response compression, reducing repeated data-fetching overhead across data-heavy workflows.
- Built 10+ reusable React components with Zustand/Redux state management and shipped 5+ Python/Django REST API endpoints with request validation, structured error handling, and consistent response formatting.
- Containerized application components with Docker and validated frontend/API behavior with Git, Postman, React DevTools, and browser debugging during Agile sprint delivery.

Wipro — Project Engineer

Mar 2021 – Jan 2022

Hyderabad, India

- Developed and maintained Java Spring Boot services and REST APIs integrating with OSI PI (PI System) real-time data feeds, supporting enterprise process-data workflows and system integrations across business operations.
- Improved OSI PI data integration stability by debugging data-sync issues and delivering targeted fixes, reducing recurring stability incidents by ~25%.
- Resolved 20+ production defects and release issues by analyzing logs and validating API behavior across environments, delivering hotfixes that cut turnaround time on customer-impacting tickets by ~30%.
- Partnered with business analysts and QA to translate requirements into scalable, maintainable backend solutions, following Git-based workflows, code reviews, and Agile delivery practices.

LVPEI — Centre for Innovation — Software Developer

Jan 2020 – Mar 2021

Hyderabad, India

- Built backend APIs and workflow automation for patient care management, EMR, and clinical data workflows at a leading eye-care institute, with strong attention to data privacy and reliability.
- Integrated a payment gateway into patient billing workflows, processing ₹25L+ in secure patient payments during the initial rollout period.
- Developed a digital patient ID system that cut average check-in processing time by ~50%, reducing manual verification steps and improving front-desk throughput across clinics.
- Integrated backend services with EMR application databases and external systems for dependable data retrieval, processing, and reporting across clinical operations.
- Translated clinical workflow requirements into secure, user-focused application modules, improving data accessibility and efficiency across internal clinical workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frontend: React.js, Next.js, Angular, Redux, Zustand, HTML5, CSS3, Tailwind, Dojo, Responsive & Component-Based UI

Backend: Python/Django, Node.js, Express.js, Spring Boot, REST APIs, OSLC, Microservices, WebSockets, Celery, Redis

Databases & Data Systems: PostgreSQL, MySQL, Oracle, DB2, Derby, MongoDB, OSI PI (PI System)

Cloud & DevOps: Azure (Microsoft Certified: Azure Data Engineer Associate – DP-203), AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD

AI / LLM Engineering: LLM Integration (IBM watsonx Granite, Anthropic Claude, OpenAI), Model Context Protocol (MCP) Servers, Tool-Based Retrieval-Grounded Workflows, Agentic Workflows, Prompt Engineering, LLM Orchestration, Tool/Function Calling, GitHub Copilot, IBM Bob

Data & ML: Pandas, NumPy, Scikit-learn, Data Analysis

Testing & Tools: JUnit, Mockito, Jest, Postman, Git/GitHub, Agile/Scrum, Code Reviews & Mentoring

Architecture: System Design, Distributed Systems, Microservices, Caching Strategies, Performance Optimization, Scalable Web Applications

EDUCATION & CERTIFICATIONS

Master of Applied Computer Science — Concordia University, Montreal, Canada 2022 – 2024

Microsoft Certified: Azure Data Engineer Associate (DP-203)